

Enhanced Traffic Monitoring using Convolutional Neural Network with Video Signal as A Labels Producer over Fiber-Optic Seismology

June, 2023

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I. ABSTRACT

Fiber-Optic Sensing is an emerging technology that allows for measurement of the seismic wave field by analyzing the signal transmitted through fiber-optic cables. One of its promising applications is urban traffic monitoring, which involves analyzing the fiber signal to gain insights into traffic patterns. However, this area of research poses significant challenges, including low Signal-to-Noise Ratio and distortions caused by heavy traffic conditions. As a result, it is difficult to extract reliable information from signals emanating from individual vehicles. Traditional methods and deep learning techniques have been explored to address these challenges, but a major obstacle has been the lack of labeled training data.

In our work, we propose a novel approach to overcome this limitation by utilizing video recordings, synchronously recording with DAS, to generate ground truth labels for training an neural network. We present a trained Convolutional Neural Network that can accurately identify and differentiate the signatures induced by traffic activity, and in particular distinguishing between cars and buses. Our methodology involves capturing video footage of urban traffic roads near which fiber optic cables were deployed, and developing an algorithmic pipeline to transform the video signal into extensive sets of ground truth labels. By integrating these labels with the fiber-optic recordings, our system can reconstruct the traffic scene solely based on the fiber-optic data, which is readily accessible in urban environments due to its widespread use for communication purposes.

The objective of our approach is to enhance the effectiveness of Neural-Networks-based traffic monitoring systems in densely populated urban environments. By leveraging the generated ground truth labels and the power of deep learning, we aim to improve the accuracy and reliability of traffic analysis and monitoring in real-world scenarios.

II. INTRODUCTION

A. Fiber Optic Seismology

Fiber optic seismology is a technique that utilizes fiber optic cables to record the seismic wave field. It involves the measurement of minuscule phase changes in the back-scattered light traveling through the cable and converting it

to directional strain along the fiber. This innovative method offers numerous advantages, such as dense spatial-temporal measurements, utilizing existing fiber infrastructure, and the ability to cover large distances with a single cable. Common applications of fiber optic seismology include earthquake detection and early warning systems, monitoring of underground reservoirs [16] and fracking operations, studying volcanic activity, and assessing structural integrity in civil engineering projects. Another application that is subject to active research is monitoring traffic in crowded areas through measuring and analyzing the signals from the fiber optic cables that are commonly deployed for communication purposes.

One significant technology that enables the sensitive and accurate measurement of the fiber-optic cables is Distributed Acoustic Sensing (DAS). By generating pulses at one terminal of the fiber and analyzing the back-scattered light from various points along the fiber, DAS technology is capable of converting standard fiber-optic cables into arrays of thousands of individual sensors [8], which are sensitive to the seismic wave-field generated by anthropogenic and natural activity around them [21].

Since fiber-optic cables for telecommunication are commonly deployed under traffic roads, it is possible to obtain a complete view of the road traffic by using DAS, utilizing existing resources without the need for additional sensors or cameras [5]. Furthermore, DAS prioritizes driver privacy, as the seismic signature of a vehicle cannot be attributed to a specific owner or model.

B. Theoretical Background

Seismic waves are elastic waves that travel through the Earth. They can result from earthquakes, landslides, explosions, surface urban activity, and more. The velocity of seismic waves depends on the elastic properties of the subsurface, its density, and type of seismic wave. Seismic waves can be categorized as surface waves, which travel along the Earth's surface, or body waves, which travel within the Earth's interior. Surface waves can be caused by various sources in urban areas. They are interface waves that travel along the free surface, but due to a skin-depth effect are influenced by the elastic properties of the subsurface. They are generally larger in amplitude and slower than body waves. They decay more

slowly with distance compared to body waves [1], as they spread cylindrically rather than spherically. The two main types of surface waves are Rayleigh waves and Love waves.

As fiber-optic cables are commonly installed underground alongside traffic roads, the weight of passing vehicles causes the subsurface to deform [14]. This deformation affects the fiber-optic cable, resulting in a tiny yet measurable strain. As the car moves away from a certain point on the road, the changes in the strain field are recorded as a strain. The seismic waves that propagate as a result of vehicular activity carry energy mainly in quasi-static frequencies which are approximately < 1 [Hz], and therefore the strain changes mainly correspond to this spectral band.

Previous works [14], [12], [4] show that the Flamant-Boussinesq approximation describes the quasi-static deformation of the subsurface caused by a theoretical point load at a given location. According to this approximation, the displacement u_x of a point x, y, z on the ground in x direction, as a response to a point load at the origin, is:

$$u_x(x, y, z) = \frac{F}{4\pi G} \frac{x}{r^2} \left(\frac{z}{r} + \frac{2\nu + 1}{1 + \frac{z}{r}} \right) \quad (1)$$

The symbols in this equation are:

- x, y, z are spatial coordinates in 3-D space in which the displacement is measured. Here we assume that the x axis is aligned with the deployment direction the fiber-optic cable, y axis is such that together with x they span the plain of the ground's surface, and z is the perpendicular depth beneath the surface.
- $r = \sqrt{x^2 + y^2 + z^2}$ is the distance between the point load location and the location of the measurement.
- F is the force applied by the point load on the ground.
- G is the shear modulus of the ground, which reflects the sensitivity of the medium to be deformed as a result of disturbance, and is assumed to be spatially uniform.
- ν is Poisson's Ratio of the medium.

C. DAS Measurements

In practical terms, the DAS system measures the strain (or strain-rate) along the fiber rather than directly measuring the displacement u_x . Additionally, the measurements are not taken at individual points along the cable, but represent an average over a certain distance known as the gauge length [14], denoted as L . Consequently, the recorded response captured by the DAS system should be modeled a spatial derivative of the average displacement along the gauge length of eq.1:

$$\varepsilon_{DAS} = \frac{1}{L} \left[u_x \left(x + \frac{L}{2}, y, z \right) - u_x \left(x - \frac{L}{2}, y, z \right) \right] \quad (2)$$

Furthermore, since the DAS system that was used in our work was optically configured to measures the strain rate, the response of the medium should actually be differentiated with respect to time as well. In conclusion, the response of the fiber strain rate to a point load at the origin, as measured by DAS system, is modeled as:

$$\dot{\varepsilon}_{DAS} = \frac{1}{L} \left[\dot{u}_x \left(x + \frac{L}{2}, y, z \right) - \dot{u}_x \left(x - \frac{L}{2}, y, z \right) \right] \quad (3)$$

The DAS measurement procedure involves setting specific temporal and spatial resolutions for sampling the fiber. This configuration results in a 2-D array output, where one axis represents time and the other axis represents distance along the fiber. Each spatial sampling point along the fiber is referred to as a channel.

D. Related Work

Previous studies have primarily focused on investigating different aspects of traffic signal analysis using distributed acoustic sensing (DAS). Some researchers have explored road-side seismic interferometry as a mean to understand traffic patterns, specifically traffic flow [10], [9]. Additionally, image processing techniques have been employed to address the spatial and temporal correlations among signals collected from each sensor within the distributed sensing array [13].

Dealing with the challenges posed by DAS data has also been a subject of research, with deep learning emerging as a potential solution. Previous works, such as [15], [20], have suggested utilizing the inherent spatial and temporal coherence of DAS data to mitigate incoherent noise, thereby enhancing data quality for subsequent tasks. Another approach proposed the use of Generative Adversarial Networks to synthesize high-quality DAS data [7], compensating for the scarcity of reliable data for neural network training.

In the context of vehicle detection and classification, a study employed acoustic frequency-based features and a Support Vector Machine (SVM) to classify individual vehicles into different groups [11]. Furthermore, deep learning techniques have been applied to extract valuable information from DAS data for traffic analysis purposes. For instance, [14], [19] attempted an unsupervised learning approach using the U-Net architecture to interpret seismic data based on seismic model equations. Another work, outlined in [17], focused on estimating vehicle occupancy from DAS measurements to enable intelligent transportation applications.

To the best of our knowledge, the development of an algorithmic flow for generating ground truth labels with the objective of training a network to distinguish between different vehicle types, such as buses and cars, remains unexplored. This research gap is a critical component in the advancement of futuristic smart traffic monitoring systems.

III. METHOD

In our research, we leverage a DAS interrogator unit to capture seismic signals within its sensitivity range. The output of the interrogator unit consists of seismic signals recorded over a specific time period, presented as 2-D arrays. These arrays exhibit spatial and temporal correlations between neighboring samples, allowing them to be interpreted as grayscale images. To extract traffic signatures from the DAS recordings, we employ a Convolutional Neural Network. The efficiency and

effectiveness of the network rely heavily on the availability of accurate ground truth labels that align with the fiber-optic signal. It ensures that the network is trained with accurate information, enabling it to effectively detect and classify traffic signatures within the DAS recordings. By aligning the ground truth labels with the fiber-optic signal, we aim to optimize the network's performance and enhance its ability to accurately analyze and interpret the traffic patterns present in the data. There are three sections in the work plan:

1) Video Camera Signal - Object Detection & Extraction: A video camera is utilized alongside the DAS interrogator to record the scene simultaneously. To extract all objects from the recorded video frames, the widely recognized YOLOv7 model [18] is applied for detection and classification. However, generating labels that align with the optic-fiber space required additional actions to be taken:

- *Removal of standing objects*: YOLO detects all objects in the frame, while the DAS is sensitive to moving objects exclusively. Thus, standing objects must be removed to ensure consistency between the two datasets.
- *Pixel-channel mapping*: A mapping is essential to establish a connection between a point (x, y) in the video frame and a single sensing point (referred to as channel) along the optic-fiber. This mapping facilitates the projection of each object detection from the video signal onto the optic-fiber space.
- *Projection*: Object detections from the video signal are projected onto the optic-fiber space based on the generated pixel-channel map. During this stage, object classification is preserved and assimilated in the new space.
- *Time calibration*: Despite steps that are taken during the data acquisition phase, further calibration efforts are required as slight time shifts can be still present. To improve label quality, a time shift is applied as needed, aligning the targets and labels more accurately.

- 2) Optic-Fiber Signal Filtering: The raw data obtained from the optic-fiber contains noise and irrelevant information. To prepare the input samples for training, F-K filtering is performed to enhance data quality and remove unnecessary information underlies in the recorded signal.
- 3) Training a Deep Neural Network: Given the complexity of the data, we employ a deep neural model to estimate the desired algorithm. Through the selection of an appropriate loss function and optimization of the model parameters, we obtain the desired output. Once trained, the network is fed by the optic-fiber data, in order to inference a clear image that accurately represents the presence of moving objects and their respective classifications.

The complete method is described in Figure 1.

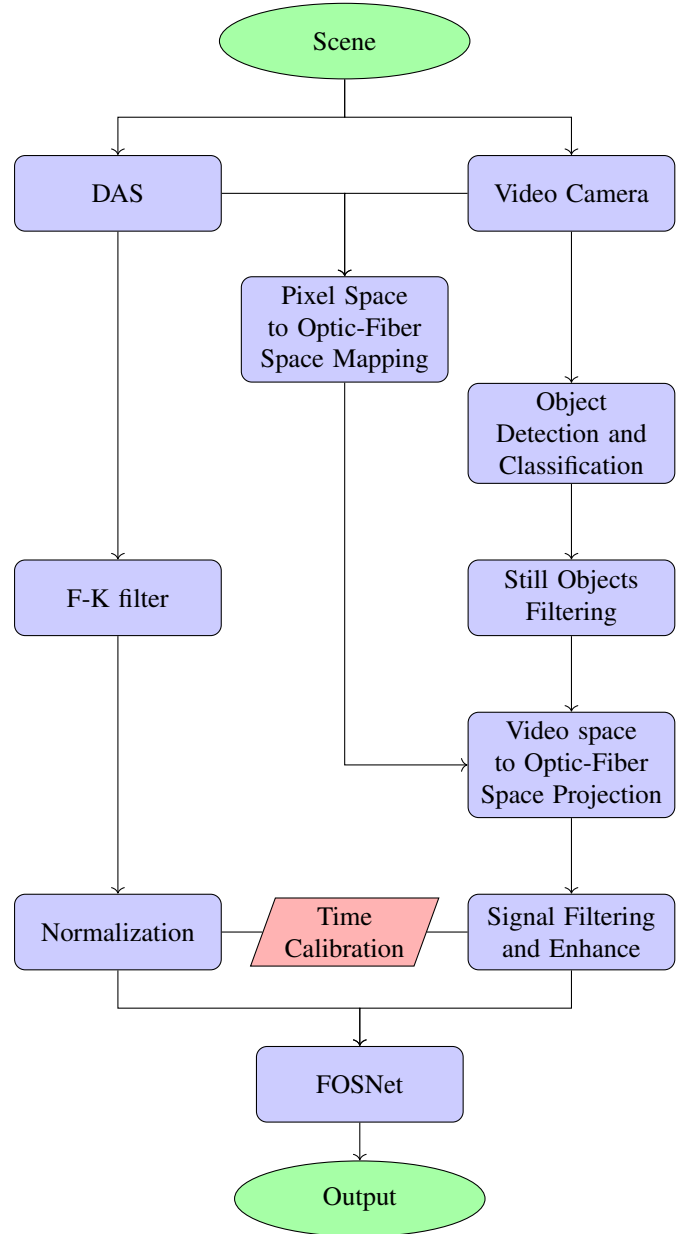


Fig. 1: A flowchart illustrating the work method.

IV. IMPLEMENTATION

A. Data Acquisition Experiment

In order to gather a substantial amount of data to train a neural network, an experiment was conducted at Tel-Aviv University. The experiment involved the installation of a DAS interrogator system on a fiber-optic cable positioned beneath a straight road with moderate traffic, including various types of vehicles such as cars, buses, motorcycles, and also pedestrians. Simultaneously, a basic smartphone video camera was placed on the rooftop of a nearby building, strategically positioned to capture the largest possible section of the monitored road. The experiment was conducted on March 21st, 2023, from 8am to 5pm.

To ensure a strong correlation between the video recording and the DAS measurements in terms of both time and location, a series of predetermined calibration steps were meticulously carried out. Before commencing the experiment, a total of 80 distinct points were strategically designated as reference anchors along a 130-meter road. The purpose of these anchors was to validate the actual response of the fiber-optic cable. Careful consideration was given to selecting points that could be effectively captured within the camera's field of view. During the execution of the experiment, with the video recording in progress and the DAS system installed, each designated point was subjected to impact by hammering, documenting the precise time and corresponding responsive channels along the fiber. Taking into consideration the static and stable positioning of the video camera throughout the experiment, a post-execution analysis was conducted to map the location of the 80 designated points in the pixel space of the camera to their corresponding responsive channels along the fiber.

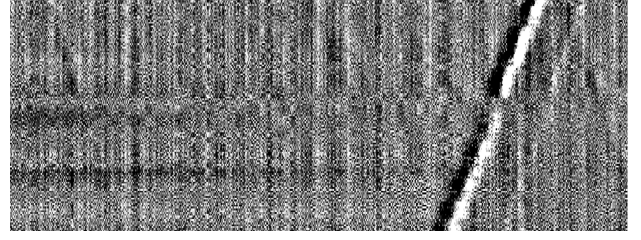
B. DAS Signals Pre-Processing

The data utilized in our research was captured on March 21st, 2023. The recorded fiber was situated adjacent to a dual carriageway accommodating various modes of transportation, including cars, buses, motorcycles, and pedestrians. Furthermore, within the recorded road segment, there exists a designated bus stop. To the best of our knowledge, the fiber's alignment closely matches the direction of the road. These descriptions, together with additional factors such as fiber-medium coupling, have a notable impact on the data. Since our primary objective was to detect and classify vehicular movements, the pre-processing steps employed were specifically tailored to facilitate this task.

As mentioned before, the data that was captured by our DAS system is shaped as a matrix with one axis corresponds with time and the other with distance. The two axes are affected by the spatial sampling rate, that was determined to be $1[m^{-1}]$, and the temporal sampling rate, that was set to $1[KHz]$. This temporal sampling rate is not necessary, as the relevant frequency band that carry vehicular movement signature are quasi-static. Thus, the data was down-sampled to $30[Hz]$, containing frequencies up to $10 [Hz]$.

In order to eliminate signals that exhibit velocities inconsistent with vehicles, while excluding any waves with frequencies higher than $3Hz$ ($2Hz$ for quasi-static waves with a safety margin), it is necessary to employ a two-dimensional filter. A one-dimensional filter alone cannot effectively address the underlying two-dimensional problem and fails to consider the signal speed along the x-axis. Thus, we utilized an F-K filter.

F-K filtering involves the conversion of seismic data, typically represented in the time and displacement domain, into the frequency and wave number (F-K) domain. The seismic data is then subjected to filtering to eliminate undesired frequencies outside the desired seismic signal range. Subsequently, the data is converted back to the time-displacement domain. In simple terms, this filter consists of a two-dimensional Fourier



(a) 1 minute DAS data before F-K Filter



(b) 1 minute DAS data after F-K Filter

Fig. 2: F-K Filter Impact On DAS Data

transform, followed by a two-dimensional band-pass filter, and finally a reverse two-dimensional Fourier transform. Our use of the F-K filter was tailored to capture the characteristics of the road that was monitored in the data acquisition experiment, hence we eliminated any signal that have wave number (K) bigger than 0.045 , and any velocity that is outside the range $[2.5, 20] [m/s]$ (corresponds with $9-72 [Km/hr]$). See Fig.2 as a visualization of the F-K Filter impact over our DAS data.

C. Labels Generation

The primary focus of our research revolves around the generation of accurate ground truth labels for neural network training using video recordings. Our objective is to create segmentation masks that align with the input DAS data, enabling pixel-level identification of noise, car signatures, and bus signatures. An essential aspect of this process involves identifying, classifying, and tracking moving vehicles on the road. To accomplish this, we employed YOLOv7 [18], an advanced object detection model, to detect and classify vehicles within the video signal (refer to Fig.3). However, the utilization of YOLO in conjunction with simple smartphone cameras introduced two challenges that impacted the accuracy and precision of label generation.

The first challenge arises from YOLOv7 frames analysis, which disregards temporal correlations between frames, leading to inconsistent object tracking throughout the video. There were instances where objects were not consistently detected in consecutive frames or were classified differently. To address this issue, classical image processing techniques were implemented in a subsequent stage to rectify missing or inconsistent detections.

The second challenge relates to the decreased detection rate of objects located farther away from the camera, resulting in the omission of vital information from the labels, particularly towards the end of the road. To mitigate this far-field miss-

detection, we only utilized half of the data from the section where the fiber-optic cable was closer to the camera, ensuring more accurate and consistent vehicle detection.

Consequently, due to the aforementioned issues present in both the YOLOv7 detection model and our recording equipment, additional corrections were necessary to ensure the satisfactory quality of the generated labels for training the network.

After processing the videos with YOLOv7 for object detection, the following steps were implemented to transform YOLOv7 detections into reliable labels:

- 1) Development of an algorithm to filter out stationary objects. For each detection, the algorithm searches backward through several frames and determines if there are any detections at the same (x, y) point, indicating that the object is stationary. This process results in a filtered data frame.
- 2) Generation of a mapping that links each point on the frame to a specific channel of the optic fiber. This mapping consists of two maps. The first map establishes the link between points on the road and specific channels along the optic fiber. This was achieved by striking the pavement along the road with a hammer at predetermined points while monitoring the influence on the seismic signal. The mapping is generated by coupling each point with the corresponding channel. The second map transforms each point on the frame to the closest point on the road from a list of points, utilizing distance calculation.
- 3) Projection of each detection from the video space to the optic fiber channel space using the established mapping. Time synchronization is based on the absolute time captured by the interrogator and the video camera. The projection process differs based on the object's class, with car detections represented in green and buses in red.
- 4) Addressing noise and imperfections introduced by the utilization of YOLOv7 in conjunction with our video recording equipment. Discontinuous lines in the images require connection, while sparse noises necessitate suppression. To connect broken lines, we employed the dilation morphological operation with a structuring element tailored to each pixel's slope. A set of predefined elements were chosen based on their correlation with each pixel, as the images contain diagonal lines in both directions. The image was split into two separate images representing positive and negative diagonals using a 2-D Fourier Transform.
- 5) To eliminate sparse noisy detections, pixels lacking a sufficient number of nonzero neighbors were nullified by convolving the image with a box filter and applying an empirically chosen threshold.
- 6) Finally, the lines were thickened to match the appearance of the DAS data, ensuring a more accurate alignment between the generated labels



Fig. 3: One frame with YOLO detection and classification.

D. Neural Network

The objective of the training process was to classify each pixel based on whether it contained a signal sample generated by the movement of cars, buses, or if it was simply noise. Given that the classification should rely on both spatial and temporal correlations between neighboring samples along the time and distance axes, an image segmentation approach was employed. To accomplish this, a U-Net architecture [2], known for its effectiveness in similar tasks, was utilized.

Our research primarily focused on detecting and classifying cars and buses, without considering other types of vehicles or the impact of pedestrians on the fiber response captured by the DAS system. Consequently, it was necessary to modify the model to incorporate a segmentation of three classes: cars, buses, and none (representing noise or other elements). This adjustment involved parameterizing the output of the model to possess three dimensions, where each dimension represents a probability map indicating the likelihood of a pixel belonging to each respective class. In other words, the three dimensions correspond to the three classes.

To interpret the classification probabilities for each pixel, a Softmax activation layer was appended at the end of the network. This activation layer normalized the sum of all values along the depth of each pixel, ensuring they added up to 1, thus providing meaningful probabilities for the classification of each pixel.

In addition to the aforementioned modifications, another adaptation made to our network involved incorporating dropout layers after each double-convolutional module, a departure from the standard U-Net architecture. The inclusion of these dropout layers aimed to promote generalization and mitigate the risk of overfitting to our training data. It is worth noting that applying the Dropout technique after a convolutional layer is less conventional compared to its typical usage after a fully-connected layer. To strike a balance between regularization and preserving valuable information, we opted for a relatively modest dropout rate of 0.2.

V. NEURAL NETWORK TRAINING

In total, our dataset comprised approximately 9 hours of video footage, which we transformed into images representing

traffic behavior along an 80-meter stretch of the road. To ensure that our model learned both the noise characteristics and the distinctive signatures induced by cars and buses on the fiber, we included additional samples of DAS recordings taken during the late hours of the night when no vehicular activity was present. In practical terms, we had a total of 1150 samples, with each sample describing 30 seconds of traffic. This amounted to 9.5 hours of data. To facilitate training, the data was shuffled and divided into batches, with each batch consisting of 16 samples. This batching approach allowed the model to optimize itself based on mean gradients, reducing the influence of individual samples. Given that our labels were not perfect, averaging their impact on the model’s learning process was crucial.

Next, we partitioned 90% of the data for training purposes, while allocating the remaining 10% for validation. Since the division of data into training and validation sets was randomized for each set of hyper-parameters, there was no need in a designated test set. The training process lasted 120 epochs, employing a learning rate of 0.0001. We implemented a Cosine Annealing regime [3], which gradually decreased the learning rate to 0 over 40 epochs, then increased it until the 80th epoch, and subsequently decreased it again until the end of training.

To optimize the network, we employed the Cross Entropy Loss function, which was made possible by incorporating the Softmax activation layer at the output of the U-Net architecture. This ensured that the output range of the network was constrained between 0 and 1. The use of Cross Entropy Loss allowed us to effectively measure the dissimilarity between the predicted probabilities and the ground truth labels.

Additionally, the optimization of the network weights was performed using the AdamW algorithm [6]. This algorithm combines the advantages of adaptive weight updates with regularization techniques to encourage better generalization and prevent overfitting. By utilizing AdamW, we aimed to strike a balance between efficiently updating the network weights based on the gradients and regularizing the model to improve its overall performance.

Figure 4 illustrates that the validation loss reaches its approximate final value after roughly 30 epochs, indicating convergence. However, upon visual inspection, it became apparent that there were further improvements achieved by the neural network beyond the slight decrease in loss values during the remaining epochs. It is suspected that these refinements are not adequately reflected in the loss term due to the imperfect nature of the labels used in training. While the loss function provides a quantitative measure of the network’s performance, it may not fully capture the nuanced refinements and improvements obtained by the network as it learns, especially when working with imperfect labels.

VI. RESULTS

Throughout the paper, it has been acknowledged that the generated labels are not pixel-accurate due to the presence

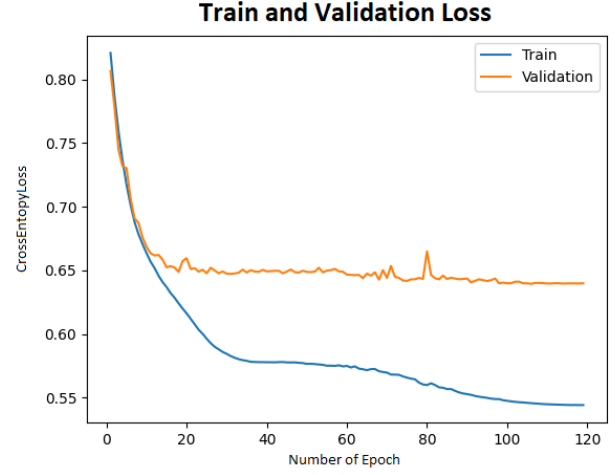


Fig. 4: Loss as Function of Training Iterations (Epochs)

of noise and imperfections in the generation pipeline. Consequently, traditional accuracy metrics are not suitable for evaluating our results. Instead, we adopted a visual measurement approach to evaluate the quality of the results. For these purposes, a subset of 20 samples was randomly selected from the validation set. Initially, we examine the accuracy of the labeling procedure, assessing how well the generated labels aligned with the ground truth as reflected from the original video recordings. Subsequently, we explored the performance of the neural network after training with these imperfect labels. Lastly, we assessed the model’s performance by comparing its output to the ground truth observed in the video. This evaluation aimed to determine whether the model effectively learned to overcome the imperfections of the labels by adapting to the average characteristics present in the ground truth data.

A. Labeling

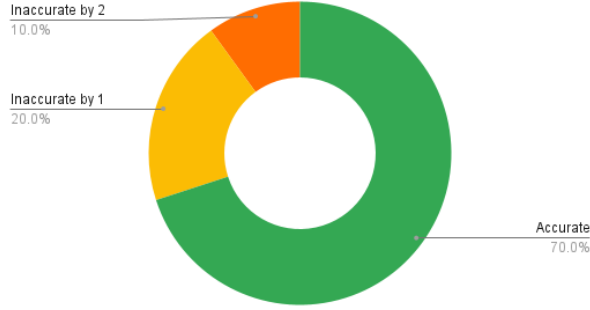
The process of generating labels involves various factors, including the imaging system configuration, camera position, occlusions in the recorded scene, and the performance of YOLOv7. These components contribute to the non-ideal nature of the labeling process, necessitating a thorough analysis of the pipeline to ensure reliable training based on these labels.

In Figure 5, we present an examination of the accuracy of the labels, categorized into two classes: buses and cars. The graphs illustrate that the labeling of buses has a higher accuracy rate of 87.5% compared to cars, which have a rate of 70%. Despite the lower accuracy rate for cars, it is worth noting that the inaccuracy is limited to at most two mislabeled cars within a 30-second timeframe.

B. Neural Network Performance

Unlike the evaluation of labeling accuracy, assessing the performance of the model required a separate analysis of the results in the two directions: northbound and southbound. Figure 10 provides a visual representation of the inputs fed

Cars Labeling Accuracy Distribution



Buses Labeling Accuracy Distribution

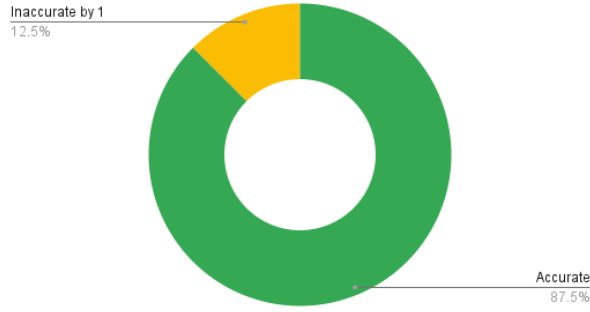


Fig. 5: Cars and Buses Labeling Accuracy Distribution. Note that inaccuracy is measured in samples of 30 seconds.

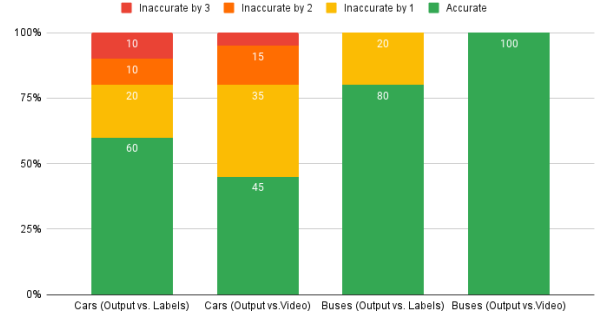
into the network, along with the corresponding labels and the resulting output. In these figures, the negatively sloped lines represent northbound movement, while the positively sloped lines represent southbound movement.

In the road where our experiment took place, the fiber is deployed closer to the northbound lane, making the signals originating from the northbound direction more significant in the data. Additionally, the northbound direction includes a bus stop, introducing additional complexities such as stopping and overtaking scenarios, which further complicate the signals. These factors contribute to the overall complexity and variability of the data that the model needs to accurately capture and classify.

Figure 6 provides insight into the accuracy of vehicle detection and classification in both the northbound and southbound directions. It is evident that buses are consistently easier to detect and classify compared to cars in both directions. However, the model's performance in accurately detecting cars in the southbound direction is relatively lower. This can be attributed to factors such as the relative distance from the fiber and the potential dominance of signals from northbound vehicles that may overshadow the signals from southbound vehicles.

Nevertheless, both Figure 6 and the qualitative results depicted in Figure 10 demonstrate significant advancements

Northbound Cars and Buses Inference Accuracy Distribution



Southbound Cars and Buses Inference Accuracy Distribution

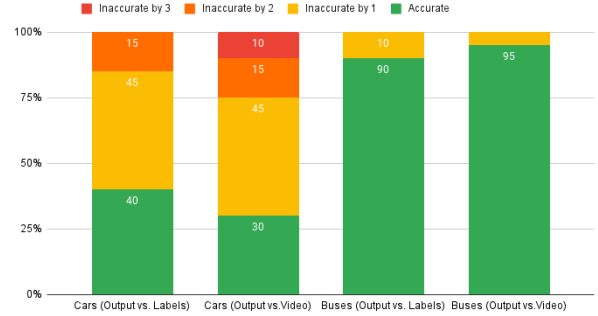


Fig. 6: Cars and Buses Inference Accuracy Distribution. Note that inaccuracy is measured in samples of 30 seconds. The two leftmost columns visualize the accuracy rate of cars detection by the model comparing the labels and the ground truth video, while the rightmost columns show the same for buses.

in the analysis of DAS data for vehicle detection and classification. These results showcase the potential of utilizing signal characteristics for accurate identification of vehicles, leading to groundbreaking outcomes in this field of research.

In order to assess the performance of the model in dealing with the imperfections of the generated labels, an accuracy measurement was conducted comparing the model's output detection to both the labels and ground truth observations from the original video. The results of this evaluation are presented in Figure 5. The findings demonstrate that the model achieved a higher accuracy rate in detecting buses compared to the labels it was trained on. This indicates that the model was able to learn and generalize from the patterns and characteristics of bus-induced seismic signals, surpassing the accuracy limitations of the imperfect labels. On the other hand, a slight decrease in accuracy rate was observed in the task of car detection. This difference is attributed to the influence of seismic signals generated by buses, as the deformation of the ground measured by the DAS system is more sensitive to the mass of the object, as described in Equation 1.

These results highlight the model's ability to effectively learn and utilize the average behavior of label imperfections,

ultimately enhancing its performance in detecting and classifying vehicles based on seismic signals captured by the DAS system.

C. Inferring Unseen Segments

Despite the challenges and limitations discussed earlier, it was observed that under certain conditions, such as moderate traffic, the results obtained from FOSNet contained sufficient information to reconstruct the traffic scene based solely on the DAS signals. However, it has been noted that the camera's position bias led to better detection and classification of objects that were closer to the camera, while objects further away were more frequently misdetected or misclassified. As a result, these further channels along the fiber were deemed unreliable for training and were excluded from the data.

After training on the more reliable segment of the road, the trained model demonstrated the ability to analyze signals captured by the DAS in the further segment. This is due to one of the key advantages of Fiber-Optic Sensing, which is its distance insensitivity. Expanding on this idea, we fed the network not just one sample of 30 seconds over approximately 70 meters, but four data samples corresponding to a whole minute and 130 meters. This approach aimed to achieve a more holistic reconstruction of the original traffic scene. The results, as shown in Figure 7, demonstrate that even in the bottom half of the middle image, which represents a road segment unseen by the network during training, the reconstructed scene remains spatially and temporally coherent.

To validate the accuracy of the results obtained for the bottom half of the scene in Figure 7, a comprehensive reconstruction and visualization of the scene was performed. The reconstructed scene was then compared against the original video that was recorded for the generation of labels. This comparison serves as a means to assess the fidelity and accuracy of the results.

To facilitate the evaluation process, a demonstration in the form of an animation was created. This animation provides a dynamic representation of the reconstructed scene and can be accessed by scanning the QR Code provided in Figure 7. By examining the animation and comparing it to the original video, researchers and readers can effectively evaluate the quality and reliability of the reconstructed scene.

VII. CONCLUSION

This research successfully achieves the identification and classification of objects from a seismic signal captured from a densely populated urban road. Moreover, it demonstrates an effective approach for generating labels with high accuracy on a large scale. By taking the seismic sensing capabilities of a dense urban road one step further, this research establishes a valuable and practical traffic monitoring tool. These findings contribute to advancing the field and opening up new possibilities for utilizing seismic signals in urban traffic monitoring applications.

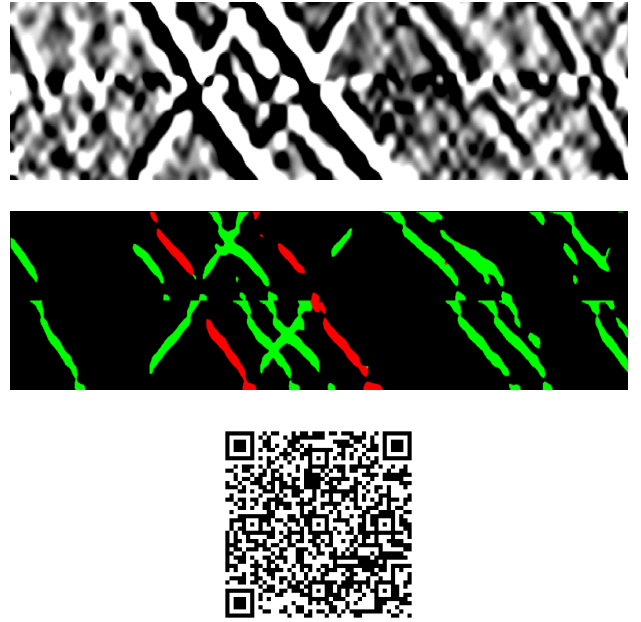


Fig. 7: Full 1-Min Scene Reconstruction: The Optic-Fiber signal input (top) and neural network's output (middle); Red color represents buses and green represents cars. The horizontal axis corresponds with duration of 1 minute, the vertical axis represents distance of 130 meters. The QR Code at the bottom provides a link to a demonstration video.

VIII. FUTURE WORK

There are two main parallel paths for further advancement in this research area, which have the potential to enhance its capabilities and applications.

The first path involves improving the accuracy of the labels. This can be achieved through various means, such as using a higher-quality camera or adjusting the camera's position relative to the road. Additionally, developing an algorithm to address the issue of fragmented lines can contribute to improving the quality of the labels. Exploring these avenues can lead to more accurate and reliable labeling of the objects in the scene.

The second path focuses on extracting additional information from the optic-fiber signal. This can be done by expanding the number of classes to encompass a wider range of objects, enabling the detection and classification of a more diverse set of entities. Furthermore, estimating the distance of objects from the fiber, determining their weight, or measuring their length are potential approaches for extracting more information from the signal. By exploring these possibilities, we can uncover valuable insights and expand the range of information that can be obtained from the optic-fiber signal.

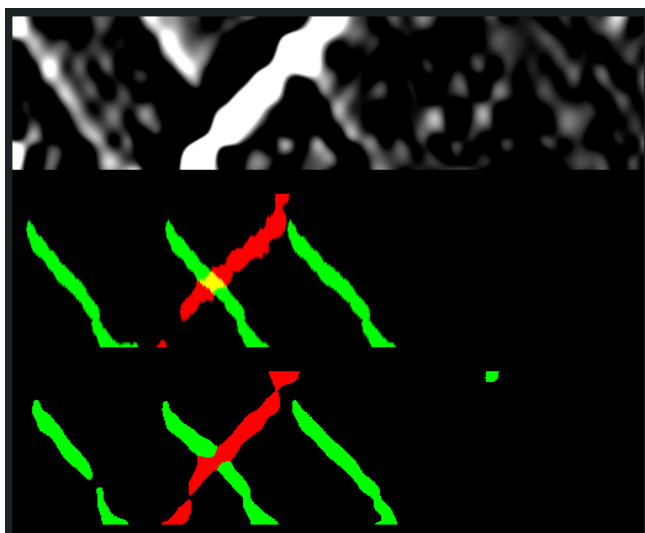
By pursuing these two paths, the full potential of this research area can be unlocked to promote further advances in its capabilities, opening up new opportunities for applications in various domains.

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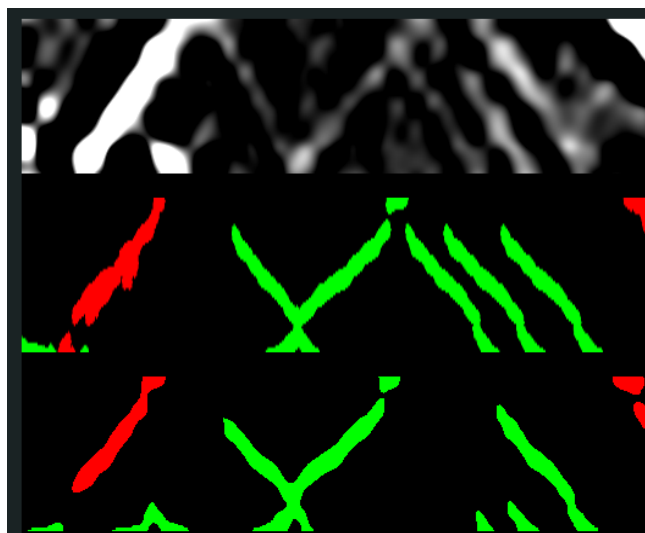
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APPENDIX

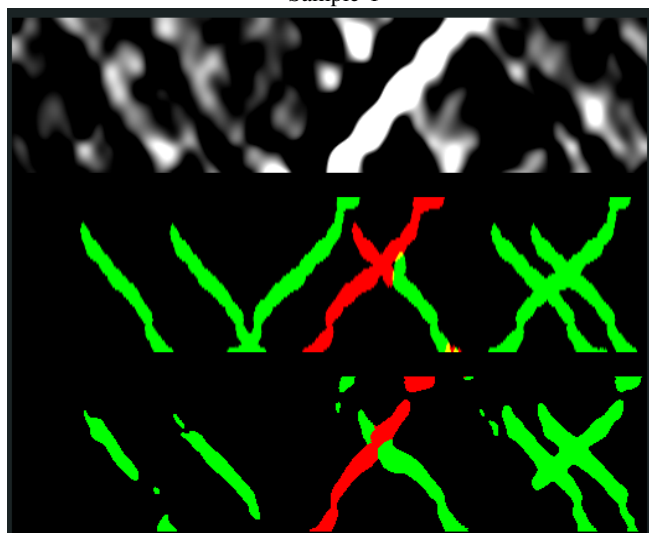
Below are examples from the validation set, consisting of triplets showcasing the input signal, the corresponding label (which was not used during training as these are samples from the validation set), and the output generated by the model for that input. These samples provide a visual representation of the model's performance, demonstrating how the input signal is processed by the model to generate the corresponding output.



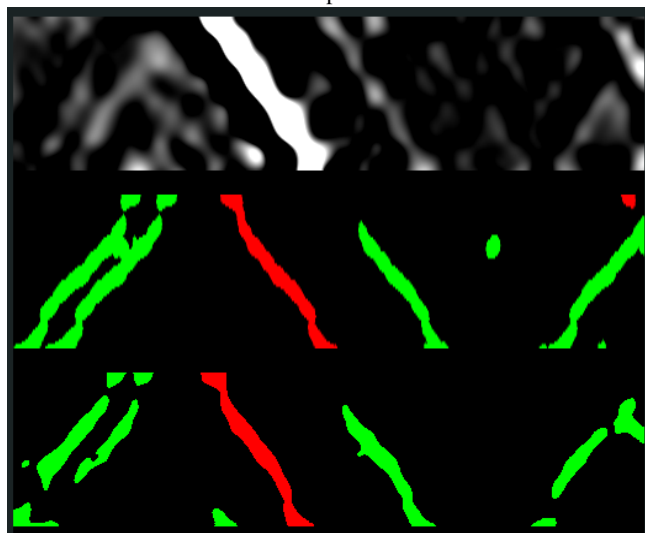
Sample 1



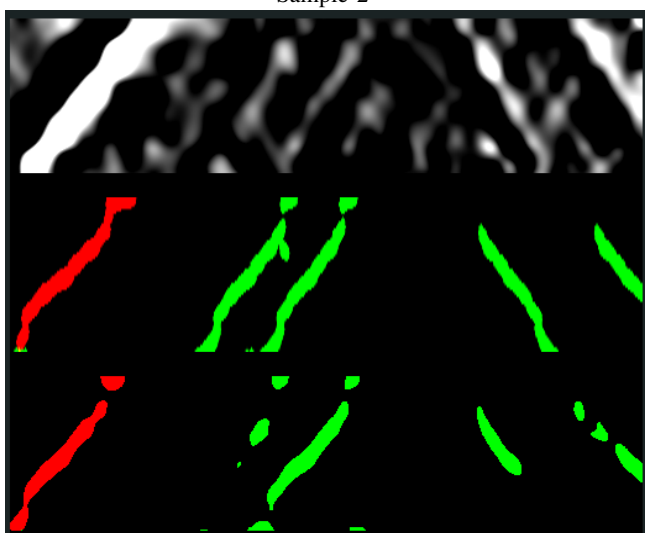
Sample 4



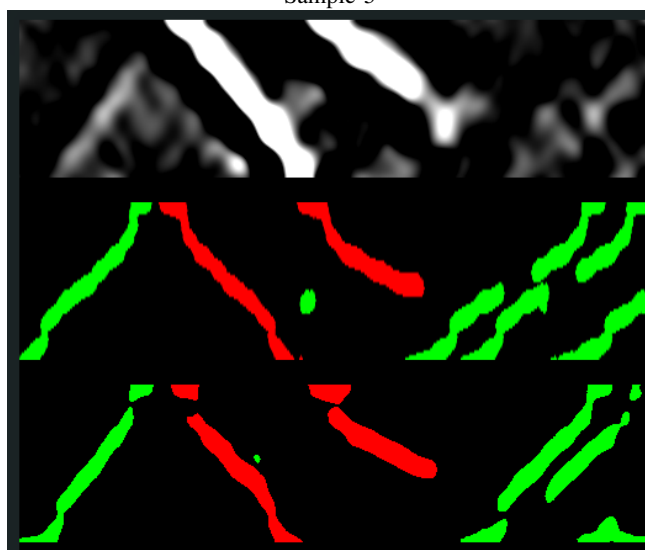
Sample 2



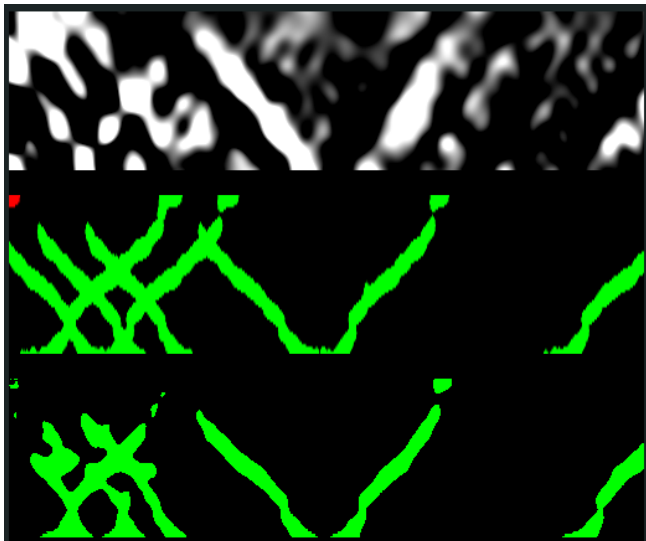
Sample 5



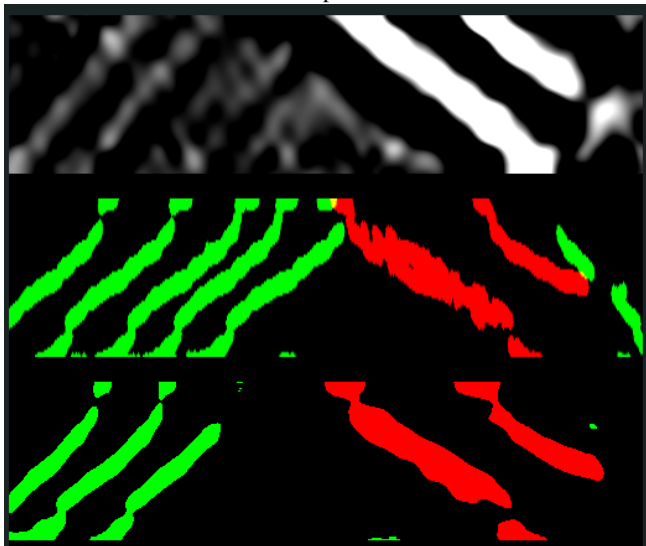
Sample 3



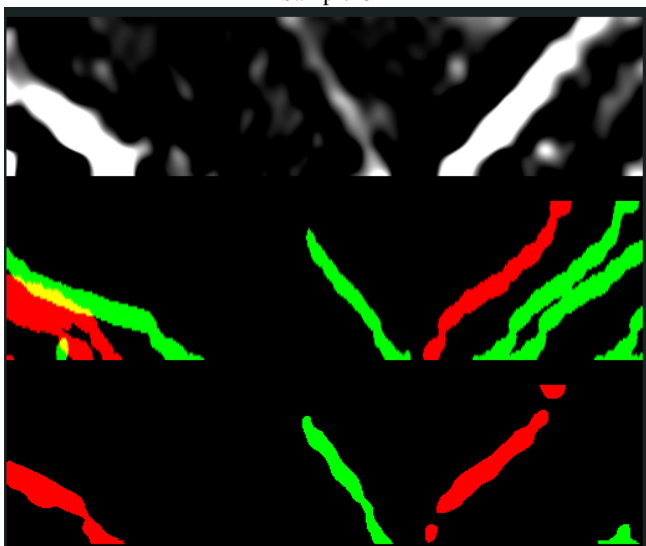
Sample 6



Sample 7



Sample 8



Sample 9

Fig. 10: The figure illustrates a series of 9 samples, each lasting 30 seconds along the X-axis. From top to bottom, the diagram depicts the input, label, and output for each sample.